Postlab

AIM:- Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

```
~
  GNU nano 8.7.1
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k = 0;
    int page_faults = 0;
    int found;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    // Initialize frames with -1
    for (i = 0; i < f; i++) {
        frames[i] = -1;
    }

    for (i = 0; i < n; i++) {
        found = 0;

        // Check if page is already in frame
        for (j = 0; j < f; j++) {
            if (frames[j] == pages[i]) {
                found = 1;
                break;
            }
        }

        // If not found → Page Fault
        if (!found) {
            frames[k] = pages[i];   // Replace using FIFO
            k = (k + 1) % f;
            page_faults++;

            printf("Page %d -> Fault\n", pages[i]);
        } else {
            printf("Page %d -> Hit\n", pages[i]);
        }
```

```
            frames[k] = pages[i];   // Replace using FIFO
            k = (k + 1) % f;
            page_faults++;

            printf("Page %d -> Fault\n", pages[i]);
        } else {
            printf("Page %d -> Hit\n", pages[i]);
        }

        // Display frames
        printf("Frames: ");
        for (j = 0; j < f; j++) {
            if (frames[j] != -1)
                printf("%d ", frames[j]);
            else
                printf("- ");
        }
        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", page_faults);

    return 0;
}
```

Output:

```
~

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ nano FIFO.c

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ gcc FIFO.c -o FIFO

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ ./FIFO
Enter number of pages: 8
Enter page reference string: 7 0 1 2 0 3 0 4
Enter number of frames: 3
Page 7 -> Fault
Frames: 7 - -
Page 0 -> Fault
Frames: 7 0 -
Page 1 -> Fault
Frames: 7 0 1
Page 2 -> Fault
Frames: 2 0 1
Page 0 -> Hit
Frames: 2 0 1
Page 3 -> Fault
Frames: 2 3 1
Page 0 -> Fault
Frames: 2 3 0
Page 4 -> Fault
Frames: 4 3 0

Total Page Faults = 7

Student@DESKTOP-JJ8CFNB MINGW64 ~
$
```